

LECTURE ONE

MATRICES

INTRODUCTION

This lecture is divided into two subtopics.

In the first subtopic, the learner is introduced to the concept of matrices. Basic definitions are given including the order of a matrix, equal matrices, row matrix, column matrix e.t.c.

The second subtopic deals with operations on matrices. These include addition, subtraction, scalar multiplication, dot product and matrix multiplication.

Each concept is illustrated by several examples. An activity in the form of an exercise is given at the end of the lecture.

Answers to the self test questions in Section 1.6 have been provided at the end of the module.

For further understanding, the learner is encouraged to read the books recommended in Section 1.5

LECTURE OBJECTIVE

By the end of this lecture, the learner should be able to perform basic operations on matrices

Including addition, subtraction, scalar multiplication and matrix multiplication.

1.1 DEFINITIONS

Matrices are used as a shorthand for keeping essential data arranged in rows and columns i.e matrices are used to summarize data in tabular form.

Definition: A matrix is an ordered rectangular array of numbers, usually enclosed in parenthesis or square brackets. Capital letters are used to denote matrices.

Order of a Matrix:

The size or order of a Matrix is specified by the number of rows and the number of columns.

A general matrix of order $m \times n$ is

$$A = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{bmatrix} = [a_{ij}] \quad i \rightarrow \text{ith row}, j \rightarrow \text{jth column}$$

A Square Matrix is a one with the same number of rows and columns i.e $m \times m$ matrix. Two matrices are of the same size if they have the same order. A vector is a matrix with one row ($1 \times n$) or one column ($n \times 1$). A row vector is of the form $1 \times n$, and a column vector is of the form $m \times 1$. A zero matrix of order $m \times n$ is the matrix with $a_{ij} = 0 \quad i = 1, \dots, m, \quad j = 1, \dots, n$. Similarly, we talk of zero rows and column vector.

$$0 = (0, 0, \dots, 0) \text{ or } \begin{bmatrix} 0 \\ 0 \\ \vdots \\ 0 \end{bmatrix}$$

Equal Matrices:

Two matrices A and B are said to be equal if they have the same order (size) $m \times n$ and $a_{ij} = b_{ij} \quad \forall i \quad \& \quad \forall j$

1.2 OPERATIONS ON MATRICES

Addition and Subtraction of matrices

This is performed on matrices of the same order (size). Let A and B be $m \times n$ matrices.

$$A + B = [a_{ij}] \pm [b_{ij}] = [a_{ij} \pm b_{ij}] = [c_{ij}] \quad (m \times n)$$

Scalar Multiplication

This is performed on any matrix and the resulting matrix is the same size

$\alpha A = \alpha [a_{ij}] = [\alpha a_{ij}]$. Each entry is multiplied by same number (scalar).

Dot product: let $\vec{a}$ and $\vec{b}$ be any two vector of size n (matrices with a single column or row).

$\vec{a} = [a_1, a_2, \dots, a_n]$, $\vec{b} = (b_1, b_2, \dots, b_n)$. The dot product of $\vec{a}$ & $\vec{b}$ denoted $\vec{a} \cdot \vec{b}$, is given by

$$\vec{a} \cdot \vec{b} = a_1 b_1 + a_2 b_2 + \dots + a_n b_n = \sum_{i=1}^n a_i b_i. \text{ The dot product is also scalar product.}$$

Dot product of a row & column vector of order n .

$$(a_1, a_2, \dots, a_n) \begin{bmatrix} b_1 \\ b_2 \\ \vdots \\ b_n \end{bmatrix} = a_1 b_1 + a_2 b_2 + \dots + a_n b_n = \sum_{i=1}^n a_i b_i = \vec{a} \cdot \vec{b}$$

Matrix Multiplication:

Let $A = [a_{ik}]$ be an $m \times n$ matrix, and $B = [b_{kj}]$ an $n \times s$ matrix. The matrix product AB is the $m \times s$ matrix $C = [c_{ij}]$ where c_{ij} the dot product of the i^{th} row of A and the j^{th} column of B .

$$\text{i.e. } AB = C, [a_{ik}][b_{kj}] = [c_{ij}] \quad ; C_{ij} = A_i \cdot B_j = \sum_{k=1}^n a_{ik} b_{kj}$$

Remark:

1. Let $A (m \times n)$, $B (s \times r)$ be two matrices .

$C = AB$ exists iff $n = s$ & C is $m \times n$

$C = BA$ exists iff $r = m$ & C is $s \times n$

2. It's possible for AB to be defined while BA is not defined. i.e. matrix multiplication is not commutative.

Examples

$$1. \text{ Let } A = \begin{pmatrix} 1 & -2 & 3 \\ 4 & 5 & -6 \end{pmatrix} \text{ and } B = \begin{pmatrix} 3 & 0 & 2 \\ -7 & 1 & 8 \end{pmatrix}. \text{ Then}$$

$$A + B = \begin{pmatrix} 1+3 & -2+0 & 3+2 \\ 4-7 & 5+1 & -6+8 \end{pmatrix} = \begin{pmatrix} 4 & -2 & 5 \\ -3 & 6 & 2 \end{pmatrix}$$

$$2. 3A = \begin{pmatrix} 3 \cdot 1 & 3 \cdot (-2) & 3 \cdot 3 \\ 3 \cdot 4 & 3 \cdot 5 & 3 \cdot (-6) \end{pmatrix} = \begin{pmatrix} 3 & -6 & 9 \\ 12 & 15 & -18 \end{pmatrix}$$

$$3. 2A - 3B = \begin{pmatrix} 2 & -4 & 6 \\ 8 & 10 & -12 \end{pmatrix} + \begin{pmatrix} -9 & 0 & -6 \\ 21 & -3 & -24 \end{pmatrix} = \begin{pmatrix} -7 & -4 & 0 \\ 29 & 7 & -36 \end{pmatrix}$$

$$4. \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix} \begin{pmatrix} 1 & 1 \\ 0 & 2 \end{pmatrix} = \begin{pmatrix} 1 \cdot 1 + 2 \cdot 0 & 1 \cdot 1 + 2 \cdot 2 \\ 3 \cdot 1 + 4 \cdot 0 & 3 \cdot 1 + 4 \cdot 2 \end{pmatrix} = \begin{pmatrix} 1 & 5 \\ 3 & 11 \end{pmatrix}$$

$$\begin{pmatrix} 1 & 1 \\ 0 & 2 \end{pmatrix} \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix} = \begin{pmatrix} 1 \cdot 1 + 1 \cdot 3 & 1 \cdot 2 + 1 \cdot 4 \\ 0 \cdot 1 + 2 \cdot 3 & 0 \cdot 2 + 2 \cdot 4 \end{pmatrix} = \begin{pmatrix} 4 & 6 \\ 6 & 8 \end{pmatrix}$$

The above example shows that matrix multiplication is not commutative, i.e. the products AB and BA of matrices need not be equal.

1.3 SUMMARY.

In this lecture we have defined a matrix, the order of a matrix, equal matrices, row matrix, column matrix and square matrix

We have also learnt how to perform addition, subtraction, scalar multiplication, dot product and multiplication of matrices.

1.4 EXERCISE 1

$$1. \text{ Let } A = \begin{pmatrix} 2 & -5 & 1 \\ 3 & 0 & -4 \end{pmatrix}, B = \begin{pmatrix} 1 & -2 & -3 \\ 0 & -1 & 5 \end{pmatrix}, C = \begin{pmatrix} 0 & 1 & -2 \\ 1 & -1 & -1 \end{pmatrix}. \text{ Find } 3A + 4B - 2C.$$

$$2. \text{ Find } x, y, z \text{ and } w \text{ if } 3 \begin{pmatrix} x & y \\ z & w \end{pmatrix} = \begin{pmatrix} x & 6 \\ -1 & 2w \end{pmatrix} + \begin{pmatrix} 4 & x+y \\ z+w & 3 \end{pmatrix}$$

$$3. \text{ Find } AB \text{ and } BA \text{ if (a) } A = \begin{pmatrix} 1 & -2 \\ -5 & 4 \end{pmatrix} B = \begin{pmatrix} -3 & -1 \\ -4 & 6 \end{pmatrix}$$

$$(b) A = \begin{pmatrix} 5 & 5 & -2 \\ 4 & 1 & -3 \end{pmatrix} B = \begin{pmatrix} 1 & 4 \\ -1 & 2 \end{pmatrix}$$

4. Let $A = \begin{pmatrix} 1 & 3 \\ 2 & -1 \end{pmatrix}$ and $B = \begin{pmatrix} 2 & 0 & -4 \\ 3 & -2 & 6 \end{pmatrix}$. Find (a) AB , (b) BA

5. Given $A = (2, -1)$ and $B = \begin{pmatrix} 1 & -2 & 0 \\ 4 & 5 & -3 \end{pmatrix}$, find (a) AB , (b) BA

6. Given $A = \begin{pmatrix} 2 & -1 \\ 1 & 0 \\ -3 & 4 \end{pmatrix}$ and $B = \begin{pmatrix} 1 & -2 & -5 \\ 3 & 4 & 0 \end{pmatrix}$, find (a) AB , (b) BA

1.5 FURTHER READING

1. Linear Algebra by Fraleigh & Beauregard
2. Linear Algebra: Schaum's Outline Series
3. Linear Algebra by J. N. Sharma, A.R. Vasishta

LECTURE TWO

PROPERTIES OF MATRIX OPERATIONS

INTRODUCTION

In this lecture we deal with the transpose of a matrix and also properties of matrix operations.

The subtopic introduces the learner to the transpose of a matrix. The definition is given, followed by a number of properties.

The second subtopic deals with properties of matrix operations. These include additive commutativity, of additive and multiplicative associativity, distributive laws etc.

Each concept is illustrated by several examples. An activity in the form of an exercise is given at the end of the lecture.

Answers to the self test questions in Section 2.6 have been provided at the end of the module.

For further understanding, the learner is encouraged to read the books recommended in Section 2.5

LECTURE OBJECTIVES

By the end of this lecture, the learner should be able to

1. Find the transpose of a given matrix.
2. Identify and verify properties of the transpose of a matrix.
3. Identify and verify properties of matrix operations like additive commutativity, additive and multiplicative associativity, distributive laws etc.

2.1 TRANSPOSE OF A MATRIX

The transpose of the matrix A is matrix $B = A^T$, such that $b_{ij} = a_{ji}$ i.e. the rows become columns and vice versa.

1. If $A = A^T$ and A is a square matrix, then we say A is symmetric. If A is symmetric,
 - a) $A A^T$ and $A + A^T$ are symmetric.
 - b) A^k symmetric $\forall_k$
 - c) If A, B are symmetric, $\alpha A + B A$ is symmetric.
2. If $A^T = -A$, we say A is skew symmetric. In that case,
 - (a) $A - A^T$ is skew symmetric
 - (b) If A, B are skew symmetric, $\alpha A + B A$ is skew symmetric.
3. If A, S are $n \times n$ (square) matrices and A is symmetric, then $S^T A S, S A S^T$ are symmetric
4. Every square matrix can be expressed as the sum of a symmetric and skew-symmetric matrix. i.e. $A = \frac{1}{2}(A + A^T) + \frac{1}{2}(A - A^T)$
 $S(A) = \frac{1}{2}(A + A^T)$ is the symmetric part; $K(A) = \frac{1}{2}(A - A^T)$ is the skew-symmetric part.
5. If A is $n \times n$ and $f(x)$ any polynomial, then $f(A^T) = [f(A)]^T$

2.2 PROPERTIES OF MATRIX OPERATIONS

1. $A + B = B + A$ - Addition is commutative
2. $(A + B) + C = A + (B + C)$ - Addition is associative
3. $A + 0 = 0 + A = A$, 0 is the identity for addition
4. $\alpha(A + B) = \alpha A + \alpha B$, left distributive law.
5. $(\alpha + \beta)A = \alpha A + \beta A$, right distributive law
6. $(\alpha \beta)A = \alpha(\beta A)$, associativity of scalar multiplication
7. $(\alpha A)B = A(\alpha \beta)$ - scalar pull through
8. $(AB)C = A(BC)$ - associativity of matrix multiplication
9. $1_n A = A$, $A I_m = A$, $A(m \times m)$ - identity for matrix multiplication
10. $A(B + C) = AB + AC$, left distributive law
11. $(A + B)C = AC + BC$, right distributive law
12. $(A^T)^T = A$
13. $(A + B)^T = A^T + B^T$ - transpose of the sum = sum of transpose
14. $(AB)^T = B^T A^T$ - transpose of product = product of transpose
 $(AB)^* = B^* A^*$.

Proof of most of these properties involve routine computations.

Show that $(A - B)(A + B) = A^2 - B^2$ iff $AB = BA$ i.e A & B commute

Solution:

$\Rightarrow$ Suppose $(A - B)(A + B) = A^2 - B^2$ and show that A & B commute.

$$\begin{aligned} A^2 - B^2 &= (A - B)(A + B) \\ &= A(A + B) - B(A + B) \\ &= A^2 + AB - BA - B^2 \end{aligned}$$

Hence $AB - BA = 0$; $AB = BA$ and so A & B commute.

$\Rightarrow$ Assume A & B commute & show that $(A - B)(A + B) = A^2 - B^2$

Since A & B commute $AB = BA$

$$\begin{aligned} (A - B)(A + B) &= A(A + B) - B(A + B) \\ &= A^2 + AB - BA - B^2 \end{aligned}$$

but $AB = BA$; $= A^2 + AB - AB - B^2 = A^2 - B^2$

2.3 SUMMARY

In this lecture, we have learnt how to find the transpose of a given matrix and to identify and verify their properties. We have also studied the properties of matrix operations including additive commutativity, additive and multiplicative associativity and distributive laws.

2.4 EXERCISE 2

1. Given $A = \begin{pmatrix} 1 & 2 \\ 3 & 5 \end{pmatrix}$, $B = \begin{pmatrix} 2 & 1 \\ 1 & 3 \end{pmatrix}$, show that $(AB)^T = B^T A^T$.

2. Let $A = \begin{pmatrix} 2 & -1 \\ 3 & 1 \end{pmatrix}$, $B = \begin{pmatrix} 1 & 2 \\ 0 & 1 \end{pmatrix}$

(a) Find A^T

(b) Calculate $A^T B A$

(c) Use your results in (b) above to determine whether or not $A^T B A$ is symmetric.

3. For what values of x , y and z is the following matrix symmetric?

$$\begin{bmatrix} x & y & z \\ 2 & 0 & 3 \\ 4 & 3 & 3 \end{bmatrix}$$

4. Find the transpose A' of the matrix $A = \begin{pmatrix} 1 & 0 & 1 & 0 \\ 2 & 3 & 4 & 5 \\ 4 & 4 & 4 & 4 \end{pmatrix}$

5. Let $A = \begin{pmatrix} 1 & 2 & 0 \\ 3 & -1 & 4 \end{pmatrix}$. Find (a) AA' , (b) $A'A$

2.5 FURTHER READING

1. Linear Algebra by Michael O'nan, Herbert Enderton
2. A First Course in Linear Algebra by Daniel Zelisky
3. Elementary Linear Algebra by Bennard Kolman
4. Elementary Linear Algebra by Howard Anto

LECTURE THREE

DETERMINANTS OF 2×2 AND 3×3 MATRICES INTRODUCTION

This lecture is divided into two subtopics.

The first subtopic covers determinants of 2×2 and 3×3 matrices. In each case, several worked out examples are given

The second subtopic deals with a general formula for finding the determinant of $n \times n$ matrices.

An activity in the form of an exercise is given at the end of the lecture.

Answers to the self test questions in Section 3.6 have been provided at the end of the module.

For further understanding, the learner is encouraged to read the books recommended in Section 3.5

LECTURE OBJECTIVES

By the end of this lecture, the learner should be able to:

1. Find the determinants of a 2×2 matrix
2. Find the determinants of a 3×3 matrix
3. Use the general formula for finding the determinant of $n \times n$ matrices to find the determinants of a 4×4 and 5×5 matrix

3.1 DETERMINANTS OF 2×2 AND 3×3 MATRICES

Let $A = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix}$. $\text{Det } A = \begin{vmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{vmatrix} = a_{11} a_{22} - a_{12} a_{21}$. The determinant is a scalar.

Examples:

1. Find the value of λ such that $\begin{vmatrix} \lambda & \lambda \\ 3 & \lambda - 2 \end{vmatrix} = 0$.

2. Let $\begin{vmatrix} 1+x & 1 \\ 2+2x & 2 \end{vmatrix} = 0$ find x .

3. Let $\begin{vmatrix} x & 3 \\ 2 & 2x+1 \end{vmatrix} = 4$. Find x .

3×3 Matrices

$$\text{Let } A = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix}$$

$$\begin{aligned} \det A &= \begin{vmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{vmatrix} = a_{11} \begin{vmatrix} a_{22} & a_{23} \\ a_{32} & a_{33} \end{vmatrix} - a_{12} \begin{vmatrix} a_{21} & a_{23} \\ a_{31} & a_{33} \end{vmatrix} + a_{13} \begin{vmatrix} a_{21} & a_{22} \\ a_{31} & a_{32} \end{vmatrix} \\ &= a_{11} (a_{22} a_{33} - a_{23} a_{32}) - a_{12} (a_{21} a_{33} - a_{23} a_{31}) + a_{13} (a_{21} a_{32} - a_{22} a_{31}) \\ &= a_{11} a_{22} a_{33} - a_{11} a_{23} a_{32} - a_{12} a_{21} a_{33} + a_{12} a_{23} a_{31} + a_{13} a_{21} a_{32} - a_{13} a_{22} a_{31} \\ &= a_{11} a_{22} a_{33} + a_{12} a_{23} a_{31} + a_{13} a_{21} a_{32} - a_{11} a_{23} a_{32} - a_{12} a_{21} a_{33} - a_{13} a_{22} a_{31} \end{aligned}$$

Note: This is a sum of **6 products**, 3 *positive* and 3 *negative*. Each product has exactly one factor from each row and column.

3.2 DETERMINANT OF AN $n \times n$ MATRIX

For a large square matrix the determinant is a sum of products, half of which have minus signs added. Each product will have exactly one factor from each row and one factor from each column.

There are $n!$ summations.

1. Evaluate $\begin{vmatrix} 3 & 1 & 2 \\ -1 & -1 & 4 \\ -2 & -1 & 1 \end{vmatrix}$

Solution: $\begin{vmatrix} 3 & 1 & 2 \\ -1 & -1 & 4 \\ -2 & -1 & 1 \end{vmatrix} = 3[(-1)(1) + 1(4)] - 1[-1(1) + 8] + 2[1(-2)] = 0$

2. Evaluate $\begin{vmatrix} 1 & 3 & 1 \\ 2 & 1 & 2 \\ 1 & -1 & 2 \end{vmatrix}$

Solution: $\begin{vmatrix} 1 & 3 & 1 \\ 2 & 1 & 2 \\ 1 & -1 & 2 \end{vmatrix} = 1 \begin{vmatrix} 1 & 2 \\ -1 & 2 \end{vmatrix} - 3 \begin{vmatrix} 2 & 2 \\ 1 & 2 \end{vmatrix} + 1 \begin{vmatrix} 2 & 1 \\ 1 & -1 \end{vmatrix} = 1(4) - 3(2) + 1(-3) = -5$

3. Evaluate $\begin{vmatrix} 1 & 2 & 3 & 1 \\ 2 & 1 & 2 & 0 \\ -1 & -2 & -1 & 1 \\ 1 & -1 & 3 & 3 \end{vmatrix}$

Solution: $\begin{vmatrix} 1 & 2 & 3 & 1 \\ 2 & 1 & 2 & 0 \\ -1 & -2 & -1 & 1 \\ 1 & -1 & 3 & 3 \end{vmatrix} = 1 \begin{vmatrix} 1 & 2 & 0 \\ -2 & -1 & 1 \\ -1 & 3 & 3 \end{vmatrix} - 2 \begin{vmatrix} 2 & 2 & 0 \\ -1 & -1 & 1 \\ 1 & 3 & 3 \end{vmatrix} + 3 \begin{vmatrix} 2 & 1 & 0 \\ -1 & -2 & 1 \\ 1 & -1 & 3 \end{vmatrix} - 1 \begin{vmatrix} 2 & 1 & 2 \\ -1 & -2 & -1 \\ 1 & -1 & 3 \end{vmatrix}$
 $= 1(4) - 2(-4) + 3(-6) - 1(-6) = 0$

3.3 SUMMARY

In this lecture, we have learnt how to find the determinants of a 2×2 matrix and a 3×3 matrix. We have also used the general formula for finding the determinant of $n \times n$ matrices to find the determinants of a 4×4 and 5×5 matrix

3.4 EXERCISE 3

1. Find the value of c such that $\begin{vmatrix} c & -3c \\ -1 & -c \end{vmatrix} = -2$

2. Find the value of λ if the matrix $A = \begin{bmatrix} \lambda & \lambda \\ 3 & \lambda & -2 \end{bmatrix}$ is singular.

3. Find $\begin{vmatrix} 2 & 3 & 4 \\ 1 & 1 & -1 \\ 4 & -1 & 2 \end{vmatrix}$

4. Determine the values of x for which the determinant of A is zero where

$$A = \begin{pmatrix} x-2 & 4 & 3 \\ 1 & x+1 & -2 \\ 0 & 0 & x-4 \end{pmatrix}$$

5. Evaluate $\begin{vmatrix} 2 & 3 & -4 \\ 0 & -4 & 2 \\ 1 & -1 & 5 \end{vmatrix}$

6. Compute the determinant of $A = \begin{pmatrix} 5 & 4 & 2 & 1 \\ 2 & 3 & 1 & -2 \\ -5 & -7 & -3 & 9 \\ 1 & -2 & -1 & 4 \end{pmatrix}$

7. Evaluate the determinant of each matrix:

(a) $\begin{pmatrix} 3 & -2 \\ 4 & 5 \end{pmatrix}$, (b) $\begin{pmatrix} a-b & a \\ b & a+b \end{pmatrix}$

(i) $\begin{vmatrix} 3 & -2 \\ 4 & 5 \end{vmatrix} = 3 \cdot 5 - (-2) \cdot 4 = 23$ (ii) $\begin{vmatrix} a-b & a \\ b & a+b \end{vmatrix} = (a-b)(a+b) - a \cdot a = -b^2$

8. Determine those values of k for which $\begin{vmatrix} k & k \\ 4 & 2k \end{vmatrix} = 0$.

$\begin{vmatrix} k & k \\ 4 & 2k \end{vmatrix} = 2k^2 - 4k = 0$, or $2k(k-2) = 0$. Hence $k = 0$; and $k = 2$. That is, if $k = 0$ or $k = 2$, the determinant is zero.

9. Compute the determinant of each matrix:

(a) $\begin{pmatrix} 1 & 2 & 3 \\ 4 & -2 & 3 \\ 2 & 5 & 1 \end{pmatrix}$ (b) $\begin{pmatrix} 2 & 0 & 1 \\ 4 & 2 & -3 \\ 5 & 3 & 1 \end{pmatrix}$ (c) $\begin{pmatrix} 2 & 0 & 1 \\ 3 & 2 & -3 \\ -1 & -3 & 5 \end{pmatrix}$

3.5 FURTHER READING

1. Linear Algebra by Fraleigh & Beauregard
2. Linear Algebra: Schaum's Outline Series
3. Elementary Linear Algebra by Howard Anton

LECTURE FOUR

PROPERTIES OF DETERMINANTS

INTRODUCTION

In this lecture, the learner is introduced to the concept of properties of determinants of matrices, which can be used to find the determinant of a matrix from the determinant of another matrix.

The concept is illustrated by several examples. An activity in the form of an exercise is given at the end of the lecture.

Answers to the self test questions in Section 4.5 have been provided at the end of the module.

For further understanding, the learner is encouraged to read the books recommended in Section 4.4

LECTURE OBJECTIVE

By the end of this lecture, the learner should be able to use properties of determinants of matrices to find the determinant of a given matrix from the determinant of another matrix.

4.1 PROPERTIES OF DETERMINANTS

1. $\text{Det } (AB) = \text{Det } A \cdot \text{Det } B$
2. If one row or column of A is multiplied by scalar r to get B , $\det(B) = r \det A$
3. If row $i = 0$ or column $j = 0$, $\det A = 0$.
4. For an $n \times n$ matrix, $\det(rA) = r^n \det A$
5. If two rows or columns are identical $\det A = 0$
6. If one row or column is a scalar multiple of another, $\det A = 0$.
7. Adding a scalar multiple of a row or a column to another row or column respectively leaves the determinant unchanged.

4.2 SUMMARY

In this lecture, we have learnt to use properties of determinants of matrices to find the determinant of a given matrix from the determinant of another matrix.

4.3 EXERCISE 4

1. (a) Evaluate $\begin{vmatrix} 2 & 3 & 4 \\ 5 & 6 & 7 \\ 8 & 9 & 1 \end{vmatrix}$ (b) Hence find $\begin{vmatrix} 18 & 21 & 6 \\ 5 & 6 & 7 \\ 8 & 9 & 1 \end{vmatrix}$

Using properties of determinants, evaluate:

2. $\begin{vmatrix} 3 & 0 \\ 2 & 1 \end{vmatrix}$ 3. $\begin{vmatrix} 2 & 1 \\ 4 & 3 \end{vmatrix}$ 4. $\begin{vmatrix} 4 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 3 \end{vmatrix}$ 5. $\begin{vmatrix} 4 & 1 & 3 \\ 2 & 3 & 0 \\ 1 & 3 & 2 \end{vmatrix}$ 6. $\begin{vmatrix} 4 & 2 & 2 & 0 \\ 2 & 0 & 0 & 0 \\ 3 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{vmatrix}$

7. $\begin{vmatrix} 4 & 2 & 3 & -4 \\ 3 & -2 & 1 & 5 \\ -2 & 0 & 1 & -3 \\ 8 & -2 & 6 & 4 \end{vmatrix}$ 8. $\begin{vmatrix} 2 & -2 \\ 3 & -1 \end{vmatrix}$ 9. $\begin{vmatrix} 4 & 2 & 0 \\ 0 & -2 & 5 \\ 0 & 0 & 3 \end{vmatrix}$ 10. $\begin{vmatrix} 3 & 4 & 2 \\ 2 & 5 & 0 \\ 3 & 0 & 0 \end{vmatrix}$ 11. $\begin{vmatrix} 4 & -3 & 5 \\ 5 & 2 & 0 \\ 2 & 0 & 4 \end{vmatrix}$

12. $\begin{vmatrix} 4 & 0 & 0 & 0 \\ -1 & 2 & 0 & 0 \\ 1 & 2 & -3 & 0 \\ 1 & 5 & 3 & 5 \end{vmatrix}$ 13. $\begin{vmatrix} 2 & 0 & 1 & 4 \\ 3 & 2 & -4 & -2 \\ 2 & 3 & -1 & 0 \\ 11 & 8 & -4 & 6 \end{vmatrix}$

14. $\begin{vmatrix} 1 & 2 & 3 \\ 2 & -1 & 3 \\ 1 & 0 & 1 \end{vmatrix}$ 15. $\begin{vmatrix} 5 & 0 & 9 \\ 2 & -1 & 3 \\ 1 & 0 & 1 \end{vmatrix}$ 16. $\begin{vmatrix} 4 & 3 & 2 \\ 3 & -2 & 5 \\ 2 & 4 & 6 \end{vmatrix}$ 17. $\begin{vmatrix} 4 & 3 & 2 \\ 3 & -2 & 5 \\ 1 & 2 & 3 \end{vmatrix}$

4.4 FURTHER READING

1. Linear Algebra by J. N. Sharma, A.R. Vasishta
2. Linear Algebra by Michael O'nan, Herbert Enderton
3. A First Course in Linear Algebra by Daniel Zelisky